HM Government

T-LEVELS

Student Workbook: Catalysts and Reaction Rate

T Level Technical Qualification in Science • Chemistry • A11.3

Name: _____ Date: _____

Specification coverage

Point	Required understanding
A11.3.1	Catalysts increase reaction rate without themselves being permanently chemically changed.
A11.3.2	Effect of catalysts in terms of the Maxwell–Boltzmann distribution.

1. Retrieval

1.1 Collision theory

Define activation energy and state the two conditions for a successful collision.

_____

_____

_____

_____

Pearson

2. Defining a catalyst

2.1 Definition

Write the complete definition of a catalyst.

_____

_____

_____

Scenario	Rate increased?	Permanently chemically changed?	Catalyst?	Justification
X makes the reaction faster and is recovered unchanged overall.				
Y makes the reaction faster but is permanently converted into a product.				
Z is recovered unchanged but has no effect on reaction rate.				

2.2 Evidence

What evidence would you need to support the claim that an unknown substance is acting as a catalyst?

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3. Alternative pathway and activation energy

3.1 Mechanism link

Explain how a catalyst increases reaction rate using the terms alternative pathway and activation energy.

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3.2 Collision link

Explain why a lower activation energy increases the proportion of successful collisions.

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4. Maxwell–Boltzmann distribution

4.1 Same temperature

At constant temperature, does adding a catalyst change the Maxwell–Boltzmann distribution? Explain.

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_____

4.2 Two activation energies

A reaction has Ea = 85 kJ mol^{-1} without a catalyst and 52 kJ mol^{-1} with a catalyst. Which Ea line should be further left on the graph, and why?

_____

4.3 Area beyond Ea

Explain what the area to the right of an Ea line represents. How does adding a catalyst affect this area?

5. Catalyst versus temperature

Change	Maxwell–Boltzmann distribution	Activation energy	Why rate increases
Increase temperature			
Add catalyst at same temperature			

6. Error Detective

6.1

Correct this statement: 'A catalyst is any substance that speeds up a reaction.'

6.2

Correct this statement: 'A catalyst gives the reacting particles more kinetic energy.'

6.3

Correct this statement: 'A catalyst changes the shape of the Maxwell–Boltzmann distribution at constant temperature.'

6.4

Correct this statement: 'A catalyst is used up during the reaction.'

7. Catalyst identification challenge

7.1 Unknown A

A reaction takes 180 s without A and 65 s with A. A is recovered after the reaction with no permanent chemical change. Is A acting as a catalyst? Justify.

7.2 Unknown B

B increases reaction rate but is completely converted into a different substance and is not regenerated. Does the evidence satisfy the definition of a catalyst? Explain.

8. Exam style application

8.1 [2]

Define a catalyst.

8.2 [2]

Explain how a catalyst increases reaction rate in terms of activation energy.

8.3 [3]

Explain how catalysed and uncatalysed pathways should be represented on one Maxwell–Boltzmann distribution at constant temperature.

8.4 [2]

Explain why the catalysed pathway gives a greater proportion of successful collisions.

9. Final knowledge check

9.1

Give the full definition of a catalyst.

9.2

What pathway does a catalyst provide?

9.3

What happens to Ea?

9.4

Does the energy distribution change at constant temperature?

9.5

What happens to the proportion with E ≥ Ea?

9.6

Why does rate increase?